GEK-45153



GE Drive Systems

INSTRUCTIONS

"RMAT"

Remote ECMS
Multiple Access Terminal

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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INTRODUCTION

The Remote Multiple Access Terminal ("RMAT") is a remote terminal that communicates with a GE Extended Crane Management System (ECMS) via a radio modem network. The RMAT Console consists of the following components: (See Figure 1)

1. Sloped operator console with writing surface (mounted on rollers).
2. GE Operator Interface Terminal.
3. Epson LX810 Serial Printer.
4. Serial Keyboard.
5. ESTeem Wireless Modem (radio).
6. Omni-Directional UHF Antenna (450-470 MHz).
7. Lightning Arrestor.

A separate "Crane End" component package is required for each crane within the RMAT network. This package consists of the following components:

1. ESTeem Wireless Modem (radio).
2. Yagi-Directional UHF Antenna (450-470 MHz).
3. Lightning Arrestor.

The RMAT system transmits in the UHF range (450-470 MHz) with a maximum output of 3.0 watts. The ESTeem Modem has a range of a few miles with a maximum of 254 cranes being accessed from one RMAT Terminal.

The ECMS computer runs a multitasking, multiuser operating system called Concurrent DOS. This operating system allows up to six independent consoles to work from the main processor. Four of these consoles are located on the main ECMS console, thus allowing two more consoles to be remotely located from the ECMS computer. The **RMAT** is an intelligent Operator Interface Terminal configured as one of these remote consoles.

The Extended Crane Management System (ECMS) monitors crane operation by examination of the Series Six™ Programmable Controller memory. Thus the ECMS screen can show the operation of the crane (devices being turned on and off), internal operation and faults in the DC300™ converter drives (drive information is transferred to the Series Six by the local area network), and operation and faults from the Genius™ I/O system (connected by the Genius Bus). The RMAT has direct access to most of the same information through the ECMS.

IMPORTANT: This Instruction Book is intended to be used in conjunction with the "Extended Crane Management System (ECMS)" Instruction Book (**GEK-85739**). It is also recommended that the user have a working knowledge of the ECMS.

™ *Series Six, DC-300, and Genius are trademarks of the General Electric Company, U.S.A.*

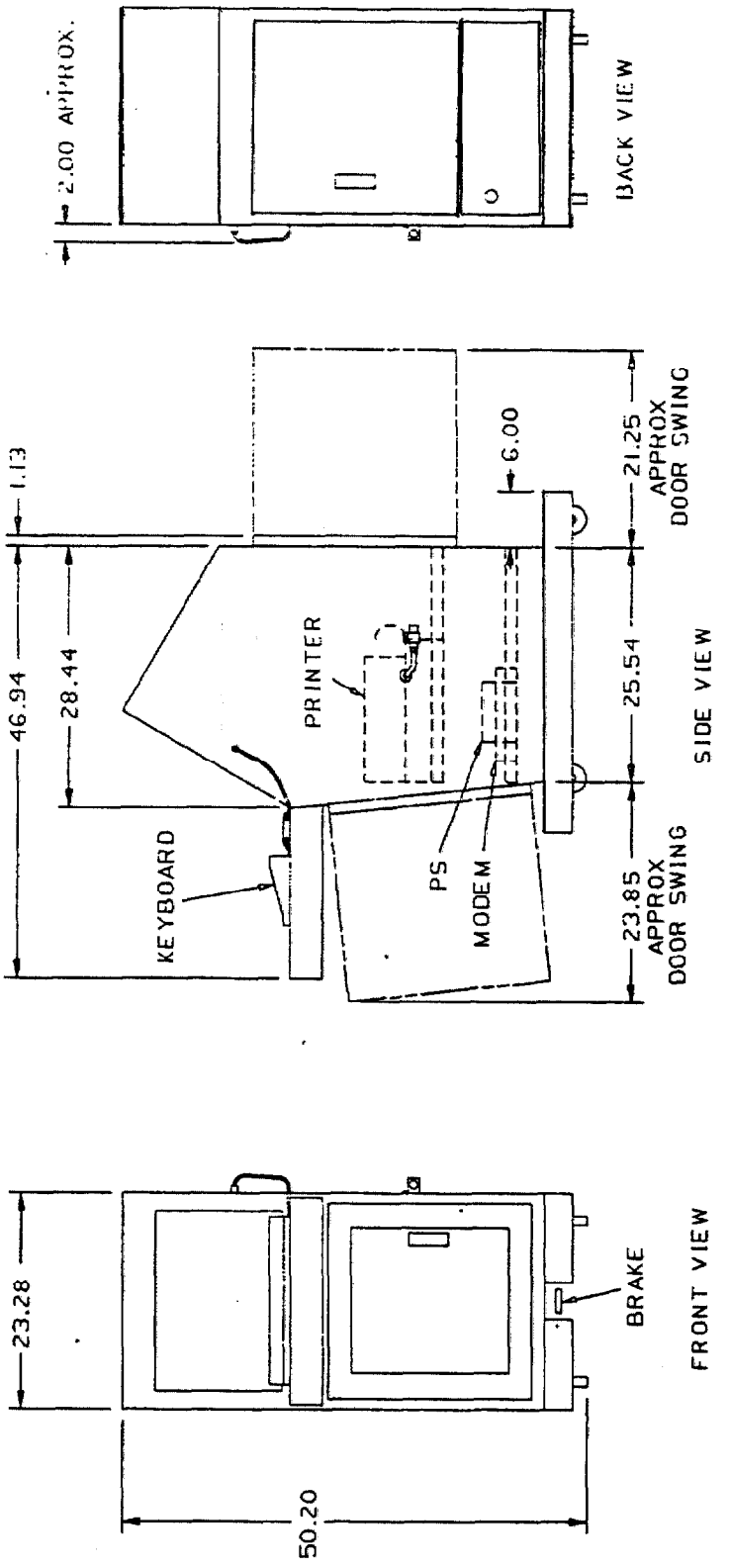


Figure 1. "RMAT" CONSOLE OUTLINE DRAWING

NOTES:

1. SAFETY PRECAUTIONS AND RECEIVING/STORAGE GUIDELINES

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used on the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.

WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.

CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.

NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when installing or operating this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - ELECTRICAL SHOCK AND ROTATING EQUIPMENT INJURY HAZARD:
Read and observe all safety precautions listed in the drive controller instruction Book (GEK) when operating or troubleshooting this equipment.

CAUTION: Do not remove connections or printed circuit cards while power is applied. This can damage the equipment.

CAUTION: Do not remove input power from the drive until it has fully executed a stop sequence as this can damage the drive system.

CAUTION: Installation wiring must be in accordance with the National Electrical Code and be consistent with all local codes.

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment and follow the start-up procedures step by step.*

NOTE: *Read and heed all **WARNING, CAUTION** and **NOTE** labels posted on the equipment.*

RECEIVING

The equipment and software should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **GE DRIVES PRODUCTS OPERATION, ERIE, PA.** When seeking assistance please use **serial number, requisition number, model number** and **drive code** to identify the equipment **(Telephone 814-875-3851).**

STORAGE

Hardware:
If this equipment is not to be installed immediately, hardware should be stored in a clean, dry location at ambient temperatures from -20° C (-4° F) to 55° C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If the storage enviroment exceeds a 15° C (27° F) drop in temperature at 50% humidity over a four hour period, a heat source should be provided to prevent condensation. (A 100 watt lamp can sometimes serve as a source of heat). Higher humidities with smaller temperature changes can also cause condensation.

Software:
Software provided on magnetic media, such as floppy disks and magnetic tape, should be stored in a clean, dry location at ambient temperatures of 10° C to 55° C (50° F to 125° F). Care should be taken to protect the media from magnetic fields, high heat or humidity and physical damage.

NOTES:

2. CONNECTING TO AN "ECMS"

PRINCIPLES OF OPERATION

As previously mentioned in the INTRODUCTION, the ECMS computer runs a multitasking, multiuser operating system. The RMAT is connected into the ECMS computer via a radio modem network operating at 4800 baud, as one of the two possible remote terminals of this multiuser system.

RMAT is configured as a console on the ECMS. Because RMAT is a serial terminal, it cannot show screens composed entirely of graphic information (such as the trendgraphs), nor can it run all of the programs contained on the ECMS computer that are specifically designed to address the main system console (the main screen on the ECMS itself). There are two programs of the ECMS computer located on drive C: and in subdirectory MMI\EXECUTE called **OITCON.CMD** and **OITLOAD.CMD**.

OITCON.CMD is the main program designed to link the remote terminal into the ECMS database, and to display the ECMS screens on that terminal. It also allows access into the alarm and event section of ECMS so that information may be displayed remotely.

OITLOAD.CMD is a program designed to convert the ECMS screen format files (named FOR##.IBM in the C:\MMI\PROJECT subdirectory) into RMAT formatted screen files (named FOR##.OIT).

OITLOAD.CMD will also download selected screen files into the RMAT memory for faster screen update.

CAUTION: As the RMAT is a remote device from the ECMS, connecting the RMAT before the ECMS is running, or connecting the RMAT during ECMS powerup, may cause the RMAT to operate incorrectly. Refer to the section on RMAT Recovery Procedures in such an event.

HARDWARE CONNECTIONS

RMAT Console

The RMAT console operates on 120 volt, 60 Hz AC. The console consist of a General Electric Operator Interface Terminal, ESTeem wireless modem with external power supply, serial printer, serial keyboard, AC switchblock, lightning arrestor, and the necesary coaxial cable to communicate from the modem to the lightning arrestor. The user is responsible for supplying the proper coaxial cable from the lightning arrestor to the antennas (this coaxial cable should have losses of no more than 5.32 db/100 meters).

The user-supplied coaxial cable is to run from the lightning arrestor (mounted on the side of the console) to the antenna. The omni-directional antenna is to be mounted on the exterior of the building on a 3/4" to 1-1/2" diameter mast. The omni-directional antenna should be mounted high enough so the directional antennas will point in its general direction.

NOTE: *Be sure to weatherproof all exterior coaxial cable connections. (See paragraph on Weatherproofing Exterior Coax Connections in Section 6 of this Instruction Book.)*

ECMS Console

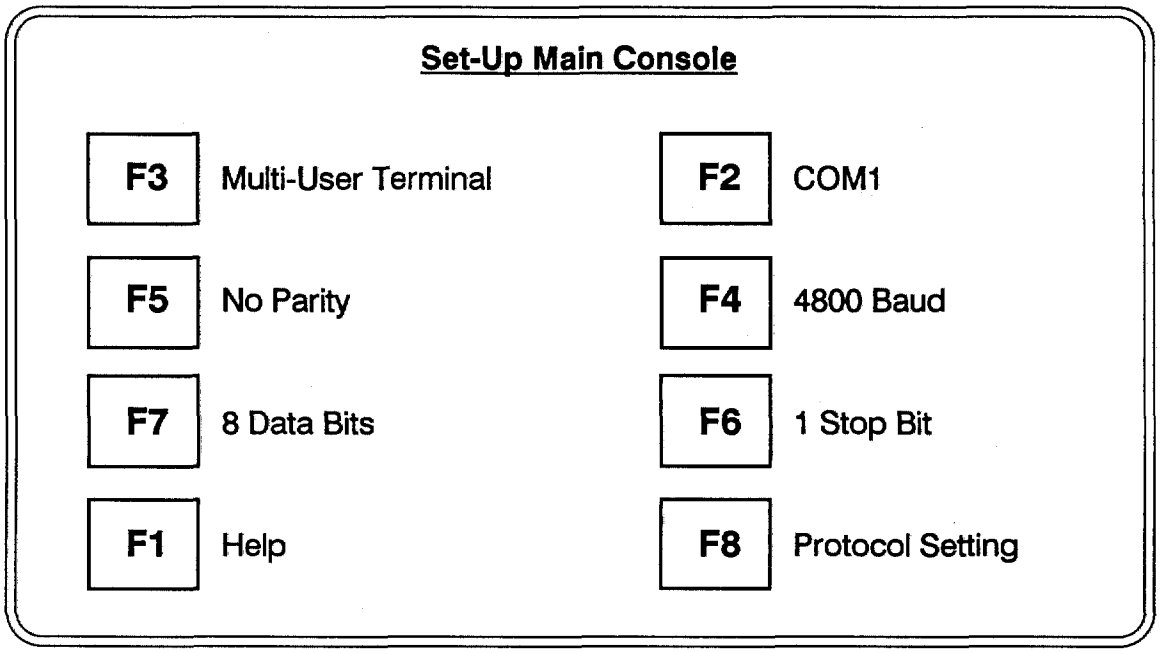
Each crane's ECMS to be connected has an ESTeem wireless modem with external power supply and a lightning arrestor installed in it. The directional antennas are to be mounted high enough on the crane to point in the general direction of the omni-directional antenna.

NOTE: *Be sure to weatherproof all exterior coaxial cable connections. (See paragraph on Weatherproofing Exterior Coax Connections in Section 6 of this Instruction Book.)*

The RS232 port of the modem is connected to COM1 of the ECMS computer. COM1 of the ECMS computer should be configured from the C: root directory. Type "SETUP" and configure the main console for 4800 Baud, No Parity, 8 Data bits, and 1 Stop bit. Be certain this set-up is saved in the root directory. (See ECMS configuration parameters shown in Figure 2-1.)

ECMS CONFIGURATION PARAMETERS

Concurrent DOS System Customization



Configure the Extended Crane Management System for RMAT by completing the following steps:

1. Create Set-Up Screen as follows:
   a. From the C:\ root directory, type "SETUP".
   b. Press <F7> for "Set-Up Workstations".
   c. Press <F3> for "Set-Up Main Console".
2. Press <ESC> twice to return to previous menu.
3. Press <F10> to update and exit.
4. Press <F5> to update system and exit.

Figure 2-1. ECMS Configuration Parameters

CONNECTING WITH AN ECMS

When power is applied to the RMAT, it performs a "self-test" of its internal systems, then assumes the "off-line" mode. The RMAT must be configured correctly for serial link speed, word length, stop bits, wrap, line feed, status lines, and type of link. These settings and how to check them are detailed in the section "RMAT Configuration". If the settings are correct the RMAT can connect to an ECMS via the radio network and become a remote terminal of that ECMS.

Normally, the RMAT and the ECMS computers will be powered up independently, i.e., the AC supply to the RMAT will be applied at a different time than the AC power to the ECMS. If RMAT attempts to connect to an ECMS that is not operating, an error message will be displayed on the RMAT screen, and no connection will be made. RMAT can only connect to an ECMS that is powered up and has completed its start-up routine.

To connect to an ECMS system press **<CONNECT>**, the number of the crane you wish to connect to, and **<ENTER>** twice. The following message will be displayed:

*****CONNECTED TO XX**

If for some reason you are unable to connect, you will see the message:

FAILED TO CONNECT

If the above message appears, you should try to reconnect. If you get the same message a second time, consider the following:

- Is power applied to the crane you are trying to connect to?
- Are you trying to connect to a valid crane number?
- Is the radio link intact?

If there is only one crane to connect to, it will be labeled 10. If there is more than one crane to connect to, they will be labeled 10, 20, 30, etc.

To load the ECMS data base into the remote terminal, press **<LOAD>** after connecting. This will run the OITLOAD program (described in previous paragraph). It will be necessary to press **<LOAD>** each time you switch to a dissimilar ECMS system.

For example, this occurs when monitoring a dockside crane, then disconnecting and trying to monitor a rubber tired crane.

- If there are any differences in the ECMS systems you are attempting to monitor, you will need to run the OITLOAD program.
- If there are no differences in the ECMS systems, you may simply disconnect from one system and connect to another.

It will take from one to two minutes to load the ECMS data base into the remote terminal. Once the ECMS data base is loaded into the remote terminal, you should press **<RUN>** to start the RMAT program. This will allow you to monitor the ECMS that you are connected to.

DISCONNECTING FROM AN ECMS

To disconnect from an ECMS, you should display the ECMS title screen by pressing function key **<F10>**, and then function key **<F4>**. You may then connect to another ECMS systerm or remain idle.

RMAT START-UP "QUICK REFERENCE GUIDE"

NOTE: *Verify that power is applied to the ECMS that you wish to connect to and that the ECMS programs are running normally.*

NOTE: *When connecting to an ECMS for the* ***first*** *time:*

a. *Press* ***<LOAD>*** *or Type* ***<LOAD>*** *on the keyboard and press* ***<ENTER>****.*
   - *The ECMS that you are connecting to will transfer its screen files to the remote terminal (this should take approximately 1 - 2 minutes).*
   - *When this has finished, you will see the prompt: C:M\MMI\PROJECT>C:\MMI\PRO.*

b. *Press* ***<ENTER>*** *to clear the command line.*

Connect to the desired ECMS by:

Pressing **<CONNECT>**, the number of the crane that you wish to connect to (10, 20, 30, etc.), and **<ENTER>** twice.

or

Typing **<CONN XX>** on the keyboard and pressing **<ENTER>** twice. (XX = 10, 20, 30, etc.)

- You will then see the following message: ***Connected to XX**

Monitor the ECMS that you are connected to by:

Pressing **<RUN>** *or* Typing **<RMAT>** on the keyboard and pressing **<ENTER>**.

- You will then see the ECMS title screen appear (this should take approximately 15 seconds).
- You are now ready to use the F-Keys to monitor crane operation.

Print the screen you are monitoring by:

Pressing **<SHIFT><PRT SC>**.

- The screen contents will be printed on the enclosed printer.

NOTE: *Occasionally erroneous data may be displayed (e.g., 23-w69 for hoist speed) while the background display remains normal. If this happens, press the F-Key labeled* ***<RESTORE>*** *(usually F1).*

Disconnect from the ECMS by:

Pressing **<F10>** from any monitoring screen, then **<F4>** to quit.

You are now ready to **<CONNECT>** to another crane or to power down the RMAT. It is only necessary to press **<LOAD>** after connecting to a different type of ECMS.

3. MONITORING SCREENS

Once the ECMS title screen is displayed on the RMAT, you will have to choose which monitoring screen is activated by pushing the appropriate function key (indicated along the top of the screen).

Function key F1 will bring up the main "System Screen" from the ECMS.

-- This screen shows all the main functions of the electrical system on the crane - AC Supply, DC Control, AC Motor Control, and Operator Inputs. This screen is designed to assist operators in diagnosing any problems they may have in operating the crane.

-- With the "System Screen" displayed, the operator has access to the individual drive screens for more detailed monitoring and diagnostics.

Pressing function key F2 from the system screen, for example, will call up the "Hoist Screen".

-- This screen has been designed to show all the functions associated with the hoist motion, and can therefore be extremely helpful in locating the cause of any malfunction in this drive.

-- On this display, the sequence of the drive flows from top to bottom - from the UV relay, to the Stop relay, to operator requests, to events within the drive, to limits that may be affecting the movement of the machine.

-- By following this natural sequence, the operator can determine the area of any trouble and either correct the problem, or notify maintenance.

Function key F6 will bring up the "Alarm and Event Console" from the ECMS.

-- This function is dealt with in the next section.

NOTE: *Pressing function key* **F10** *while viewing any of the monitoring screens will return the operator to the ECMS title screen.*

MONITORING THE SERIES SIX

It is possible to monitor individual Series Six I/O and registers using RMAT. These screens are retrieved by pressing the "MORE" function key (F8) while displaying a monitoring screen. This function is specifically for maintenance personnel only. The user must enter a valid Series Six address into the entry fields on the screens to obtain on-line data.

-- Entry fields are accessed by pressing the "DOWN ARROW" key on the RMAT.

-- Once the cursor is at the desired data field, press the "CHANGE VALUE" function key, enter the required address, then answer "Y" or "N" as needed.

NOTES:

4. EVENT CONSOLE OPERATION

"LOG" DESCRIPTION

The ECMS will keep a running log of the last 1000 alarms and events that occurred on the crane. When the system was configured for this crane, some contacts changing state (eg. from open to closed) were configured as "events" to be logged, some were configured as "alarms" to be logged, and some were for indication on the monitoring screens only. Those contacts configured as "events" will appear in the Event Log. Those contacts configured as "alarms" will appear in the Alarm Log and the Event Log.

This "log" is like a long list that has been joined at both ends to make an endless loop 1000 units long. As each alarm and event occurs, it is added to the list until one thousand have been added. At this point, the first entry is deleted to make room for the 1001 entry, and so it continues. The list therefore contains the last 1000 alarms and events that occurred on the crane. Alarms and events occuring before that are gone.

Alarms and events can occur for a variety of reasons in the system during normal operation. All of these alarms and events are translated into a message 'text' that is 'sent' to the alarm console program. The alarm console manager program provides a variety of user commands using function keys to display these 'text' messages.

ACTIVE ALARM DISPLAY

Active alarms are those messages that have either a priority of urgent or caution. When an urgent or caution alarm (let's call them high priority alarms) occurs, it is always shown on the active alarm display. When it returns to normal, it is removed from the active alarm display.

The event console manager can retain up to 100 high priority alarms for this display. When more than 100 high priority alarms exist, the oldest alarms are removed from the list.

The top of the display screen shows the available functions. Each function is selected by pressing the appropriate function key. The bottom of the display shows the current event console display at the left side and the top event number on the right side.

The top alarm is a number representing a pointer into the list of 100 high priority alarms. As you scroll through the list, the top alarm number changes to indicate where you are in the list.

Several keys are available to look at the active alarms. The **PGUP (F11)** and **PGDN (F12)** keys cause the display to page through a screenful at a time. The carriage return key scrolls up one line at a time. The up and down cursor keys allow moving up and down one line at a time. The **HOME** key brings you back to the top alarm.

The function keys' operations are described below:

Active Alarms
This function will change the display to the Active Alarm display.

Event Subset
This function will display all alarms and events that fulfil the search conditions specified in the "Load Subset" system. (See later).

Previous Page
This function will cause the screen to display the previous 22 active alarms. This function is inactive when the display is already at the top alarm number 1.

Next Page
This function will cause the screen to display the next 22 active alarms. This function is inactive when the display is already at the last alarm (#100).

Latest Alarms
This function brings the first display back up and sets the TAB bar to the latest alarm on the list. This function key should always be used first when looking at the event console.

Graph Event
This function is not used by RMAT.

Return to System
This function causes the RMAT to return to the previous monitoring screen.

More
This function key will access the next level of functions available. They appear below.

Load Subset
This function is used to interrogate the event history for specific alarm/event occurences. (See Subset functions later).

Alarm Totals
This function will total all occurences of all alarms contained in the history files.

Print Events
This function causes a printout of selected events on the RMAT printer.

Help
This function provides a help screen for the alarm/event highlighted by the TAB bar.

Acknowledge Alarms
This function is not supported by RMAT.

Subset Functions
These functions allow the user to select a subset of the total history file to be displayed. The functions vary depending on the "tabbed" alarm highlighted when the "load subset" function key is pressed. The subset chosen can be a function of alarm type, and name (use question marks ? for a wildcard symbol in a name mask).

EVENT HISTORY DISPLAY

This display is used to show all of the system events and alarms including return to normal messages. The console can display the 200 most recent alarms and events and also records the last 1000 messages to a disk file on the ECMS. When more than 200 messages exist, the oldest messages are removed from memory. When more than 1000 messages have been stored to disk, the oldest are discarded.

The top of the display screen shows the available functions to the operator. Each function is selected by pressing the appropriate function key. The bottom of the display shows the current alarm console display at the left side and the bottom alarm number on the right side.

The bottom alarm is a number representing a pointer into the list of 200 alarms and events. As you scroll through the list, the bottom alarm number changes to indicate where you are in the list.

Several keys are available to look at the event history. The **PGUP (F11)** and **PGDN (F12)** keys cause the display to page through a screenful at a time. The carriage return key scrolls up one line at a time. The up and down cursor keys allow moving up and down one line at a time. The **HOME** key brings you back to the bottom alarm.

The function keys are described below:

While most of these functions have been previously described several new ones are provided in this menu. These are described in the following sections.

Active Alarms
This function changes the display back to the active alarm display.

Latest Events
This function sets the bottom alarm back to 200 so that the 22 most recent events are shown on the screen.

ALARM MESSAGE FORMAT(S)

This paragraph is provided to help you understand the information that is contained in each of the various types of alarm and event messages. Each category of alarm or event in the system has a fixed format. This section allows you to identify what type of message format is being used and how to decode the various fields in the message.

Each alarm begins with the time and date that the alarm message text was generated. This is usually within 1 second of the actual alarm detection.

The next two to four characters define the general type of alarm that is being shown.

The possible codes are listed below:

Type	Code	Description
1	OK	Return to normal message
2	DA	Digital Alarm
3	DK	Digital Acknowledge (reserved for future use)
4	HI	High Alarm
5	HK	High Alarm Acknowledge (reserved for future use)
6	LO	Low Alarm
7	LK	Low Alarm Acknowledge (reserved for future use)
8	MDCH	Drive Mode Change Alarm
9	SPCH	Setpoint change message
10	GENI	Genius Alarm
11	##	Drive fault or ECMS Program alarm

Type 11 Alarms use a two digit number in conjunction with the text string following it to indicate what drive or ECMS program has generated the fault message.

Types 1, 8, 9, and 10 Alarms are considered events and are not displayed on the active alarm summary display. This is due to the fact that no return to normal condition for these alarms is available.

The following describes each alarm category:

Digital Alarms

```
15:20:38   03Feb88   DA U 272
COMFAIL  COMMUNICATIONS  STALLD
```

The above alarm format is used for digital alarms. Digital alarms are used for indicating faults when one state indicates normal status and the other is a fault condition. The alarm priority is indicated by the 'U' character. In this example, the priority is 'Urgent'. The number of the alarm is 272. This number is internal to the ECMS and has no real user value. The 'COMFAIL' text is a 1 to 9 character nickname (tagname) that is unique for every alarm point in the system. It is followed by an english description of the alarm point and its new state. In this case, the state is 'STALLD'. When the alarm returns to normal, the state would probably indicate 'HEALTY'.

Analog Alarms

Analog alarms are similar to digital alarms except that they provide the current value of the point when the alarm text message was created. In addition, the engineering units for the point are shown immediately following the current value.

Genius Alarms

Genius alarms are always indicated by the word 'GENIUS' in the description field. Genius alarms can indicate faults for bus controller(s), blocks, and points.

Drive Alarms

```
15:27:00   03Feb88   4   U HOIST
LOSS OF AN AC PHASE      F22
```

The drive drop number on the LAN is indicated by the number 4. The description of the alarm includes an english description of the drive along with an expanded fault description. The fault number is shown at the end of the message.

Drive Mode Change Alarms

This function is not supported in ECMS or RMAT.

Setpoint Change Messages

```
15:25:06   03Feb88   SPCH PA01SET
PARISON POINT #01    0.0000 to 56.00 %
15:26:01   03Feb88   SPCH E1MXTSP
EXT 1 M-PIPE MELT    0.0000 to 230.DEG
```

Whenever a point is changed by the operator using the CHANGE POINT function key, an english message alarm is generated. This message records the setpoint change showing the nickname (tagname), its description, the old value, and the new value with its engineering units (if applicable).

ECMS Program Alarms

```
15:23:49   03Feb88   72  E SC-SETUP
Execution from DISK complete      F93
15:27:00   03Feb88   75  E RECORDER
Pen channel started and sampling  F203
```

Several of the programs in the ECMS system generate alarm or event messages to indicate system conditions. For example, when a new pen is connected in the recorder system, an event message is generated indicating that a pen channel has started. When recipe operations complete, an event message is generated to indicate the request has been processed.

ALARM HELP

By "tabbing" to a particular alarm or event (by using the arrow keys), and pressing function key F10, on-line help will be offered by the system, describing the cause, location, and possible solutions for the specific alarm/event.

PRINTING

To print the terminals screen contents to the printer, press the **<PRINT SCREEN>** key on the keyboard.

NOTES:

5. RMAT CONFIGURATION

CONFIGURE THE OPERATOR INTERFACE TERMINAL (OIT) OF THE RMAT

In order to configure an RMAT, the following steps must be completed:

1. Turn on power to RMAT.
2. Press Ctrl 1 to call up the set-up system after RMAT has completed self test.
3. Configure the OIT to the parameters shown in the chart below.

OIT CONFIGURATION PARAMETERS

Parameter	Configuration Description					
Mode of Operation	Online	ANSI	No Echo		Send any case	
Date/Time						
End Line / Color	No Wrap		No Auto LF		Color	
Cursor/Screen	Blinking		Underline		CRT Saver On	
Status Lines	One					
Keyboard Codes	Five					
Serial Port 1 (Primary)*	4800	None	8 Bit	1 Stop	Soft	Pt to Pt
Serial Port 2 (Secondary)**	4800	None	8 Bit	1 Stop	Soft	Pt to Pt
Serial Port 3 (Tertiary)**	4800	None	8 Bit	1 Stop	None	
Configuration	Save					
Confidence Tests	Diagnostic					
Extended Memory	Save	Screens 00		to	Screens	00

NOTE: *Do not forget to save the configuration by highlighting the "Save" option and pressing F6.*

All three serial ports **must be the same baud rate.*

***Prior to **12/1/92**, Serial Port 2 and Serial Port 3 were not used.*

NOTES:

6. ESTeem MODEM INSTALLATION AND USAGE REQUIREMENTS

ESTeem MODEM SETUP

Please refer to the ESTeem User's Manual for Models 85 and 96C (included in Custom Instruction Book). All attributes of the modems are pre-programmed by the manufacturer.

NOTE: *The only reason to program the modems would be when they are replaced.* ***GE Drives Products Operation*** *should be consulted before attempting to reprogram.*

NOTE: *To insure compliance to FCC Non-Interference Regulations, peripherals attached to this modem require shielded I/O cables.*

There is an eight position switch labeled RS-232 SETUP. The switch positions should be set as follows:

ECMS End Switch Position							
1	2	3	4	5	6	7	8
Off	On	Off	Off	Off	On	On	Off

RMAT Terminal End Switch Position							
1	2	3	4	5	6	7	8
Off	On	Off	Off	Off	On	On	On

FEDERAL LICENSING REQUIREMENTS

It is illegal to operate any transmitter without the appropriate Federal Communications Commission license. Before you operate the ESTeem Modem, you must obtain a license from the Federal Communications Commisssion. The modem is licensed under Form 574. FCC Rules and Regulations Volume 5 for Part 90 will assist you in this matter.

Listed below are items to help in filling out Form 574:

1. Frequency - 450-470 MHz.
2. Emissions Designator - USA, 16KOF2O; Canada, 16KOF2DCN.
3. Output Power - 0.5 Watts.
4. E.R.P. - 0.5 Watts.
5. FCC Rules - Class A, Part 90, Subpart K.

Application forms for the required license are available from the nearest office of the Federal Communications Commission.

ANTENNAS

Attention should be paid to the installation of the antennas. The Model 85 and 96C Modems operate in the 450-470 MHz range. Follow all directions on the coaxial connectors (in particular, waterproofing, grounding, and lightning arrestors).

WARNING - BURN HAZARD: Do not operate the modem when anyone is touching the antenna or when anyone is standing within 2-3 feet of the antenna (to avoid the possibility of radio frequency burns or related injury).

WEATHERPROOFING EXTERIOR COAXIAL CONNECTIONS

1. Coat the threads of the connectors with silicone lubricant prior to assembly (see Note 1) and hand tighten.
   - Care should be taken not to get lubricant on the center conductor.
2. Wrap the connector assembly with a vapor barrier patch for weatherproofing (see Note 2).
   - Be sure to overlap onto the coaxial cable approximately 1-1/2 inches.
3. Apply an electrical coating (sealing agent) over the vapor barrier patch for added protection (see Note 3).

NOTES:

1. *Dow Corning RTV-3140 or equivalent.*
2. *Suggested Suppliers:*
   a. *Vapor-Wrap*
   Decibel Products
   3184 Quebec Street
   Dallas, Texas 75356
   Phone: (214) 631-3100
   b. *Vinyl-Mastic, P/N 2200*
   3-M Company
   Customer Service
   Phone: (512) 984-1800
3. *Scotchkote, 3-M Company or equivalent.*

COMMUNICATIONS INTERFERENCE INFORMATION

The ESTeem Modems generate, use, and can radiate radio frequency energy. If they are not installed and used in accordance with their Instruction Manual, they may cause interference to radio communications. They have been tested and found to comply with the limits for a Class A computing device pursuant to Subpart J of Part 15 of FCC Regulations that are designed to provide reasonable protection against such interference when operated in a commercial environment.

To insure compliance to FCC Non-Interference Regulations, peripherals attached to the modems require shielded I/O cables.

If the equipment does cause interference (this can be determined by turning the equipment off and then on), the user is encouraged to try to correct the interference by one or more of the following measures*:

1. Re-orient the receiving antenna.
2. Re-orient the modem antenna.
3. Relocate the modem with respect the receiving antenna.
4. Verify that power supply for modem and receiver (AC outlets)are on different branch circuits.
5. Verify that the modem chassis is connected to an earth ground.
6. Attach a split bead (such as FAIR-RITE PN#2643164251) to the RS232 cable.

**The user may also find the following booklet prepared by the FCC helpful: "How to Identify and Resolve Radio-TV Interference Problems"*
This booklet is available from the U.S. Government Printing Office, Washington, D.C. - Stock Number 004-000-00245-4.

NOTES:

7. TROUBLESHOOTING PROCEDURES

CORRUPTED TERMINAL RECOVERY

If the RMAT terminal should become corrupted, press Function Key **<F10>** to clear the screen and return to the ECMS title screen.

Each monitoring screen also has a **RESTORE** Function Key that will clear the screen and return to the screen that became corrupted.

FUNCTION KEY ONLY DISPLAY

If the RMAT terminal displays function keys only, with the rest of the terminal blank, it will be necessary to reload the ECMS data base into the remote terminal as follows.

1. Verify RMAT is connected to the ECMS.
   - -- If not connected, refer to Section 2 of this instruction.
2. Press **<LOAD>** to reload the ECMS data base into the remote terminal.
   - -- This will run the OITLOAD program .
   - -- It will be necessary to press **<LOAD>** each time you switch to a different type ECMS system.

For example, this occurs when monitoring a dockside crane, then disconnecting and trying to monitor a rubber tired crane.

- -- If there are any differences in the ECMS systems you are attempting to monitor, you will need to run the OITLOAD program.
- -- If there are no differences in the ECMS systems, you may simply disconnect from one system and connect to another.

It will take from one to two minutes to load the ECMS data base into the remote terminal. Once the ECMS data base is loaded into the remote terminal, you should press **<RUN>** to restart the RMAT program. This will allow you to monitor the ECMS that you are connected to.

ADDITIONAL TROUBLESHOOTING NOTES

1. If there has been an orderly shutdown of the RMAT monitoring operations, then the normal start-up procedure (using F-Keys) may be used to restart remote monitoring. (See first option of procedures in "Quick Reference Guide" of Section 2.)

2. If RMAT shutdown was normal but the ECMS has been reset, then the alternative procedure (using keyboard) will be necessary. (See second option of procedures in "Quick Reference Guide" of Section 2.)

3. When the display LOAD sequence is in progress and blank areas appear in one of the displays, the LOAD sequence should be restarted (upon completion of the first LOAD sequence).

4. When monitoring crane operation, if blank areas or shifting data appear on a display, there are errors in the display files held by the RMAT. The RMAT should be shutdown normally (not necessary to cycle power) and the LOAD sequence followed again.

5. If after connecting to the modem on the crane (i.e., getting the "***CONNECTED TO XX" message) and after pressing <ENTER> again there is no prompt, it means the ECMS computer is not powered up or that it is not running Concurrent DOS.

ERROR MESSAGES

1. The message "INVALID COMMAND" comes from the modem and indicates that the F-Keys are not active and that the keyboard must be used. (It may also mean that the ECMS computer that you have connected to is not powered up or that it is not running Concurrent DOS.)

2. The message "UNIT ADDRESS BE 23592**" indicates the RMAT modem has been reset or powered on after the RMAT display unit (GE Operator Interface Terminal, OIT) has completed its self-test. It is normally not necessary to reset the modem.

3. The message "LINK IS DOWN" indicates a failure in the hardware communication link or an obstruction to the line-of-sight between the antennas. This message can be received after connection has been established.

4. The message "FAILED TO CONNECT" may mean that an incorrect address was entered (try to connect again) or that the modem with this address is not powered up.

NOTES:

8. WARRANTY PROCEDURES and RENEWAL PARTS

1. Consult the **GE DRIVES PRODUCTS OPERATION Terms and Conditions** for details of the product warranty, including warranty period and parts or service coverage.

2. Warranty replacement parts or service assistance may be obtained direct from the factory during regular working hours (8 a.m. - 5 p.m.; M-F) by calling:

   GE DRIVES PRODUCTS OPERATION
   PRODUCT SERVICE ENGINEERING
   ERIE, PA
   (814) 875-3851

3. An **EMERGENCY HOT-LINE** has been established by **GE DRIVES PRODUCTS OPERATION** to provide a 24-hour telephone answering service to our customers:

   GE DRIVES PRODUCTS OPERATION
   EMERGENCY HOT-LINE
   (814) 455-3219

   For a **PROMPT RESPONSE THE NEXT WORKING DAY**, the answering service operator will record the required information and your **NAME, COMPANY,** and **PHONE NUMBER.** In cases of emergency, where assistance must be obtained immediately, the operator may be requested to contact a Customer Service Representative (for parts) or a Product Service Engineer (for technical assistance) at home.

4. So that we may serve you more expediently, please have the following information ready whenever contacting the factory:

   EQUIPMENT SERIAL NUMBER & MODEL NUMBER
   PART NUMBER & DESCRIPTION

5. **GE DRIVES PRODUCTS OPERATION** may request that failed parts replaced under warranty be returned to the factory. A **RETURN AUTHORIZATION** tag or number is to be obtained and included with any parts being returned. When returning parts, pack them carefully to prevent damage in transit and ship to:

   GENERAL ELECTRIC COMPANY
   DRIVES PRODUCTS OPERATION
   1100 LAWRENCE PARKWAY
   ERIE, PA 16531
   ATTENTION: Parts Repair and Exchange

SPARE/RENEWAL PARTS

Spare and/or Renewal Parts can be obtained for the "RMAT" and related equipment by contacting Product Service and referencing the following part numbers.

NOTE: *For the ESTeem Modems, refer to the manufacturers "Users Manual" for parts listing. Refer to* ***GEK-90817*** *for parts listing for the OIT.*

"RMAT" Console Replacement Parts

Description	Part Number
Elementary Diagram	255B2612BA001
Elementary Diagram	255B2612BA002
RMAT Console	339A1130AA001
Printer	339A1130AA002
Wireless Modem	339A1130AA003
Power Supply	339A1130AA004
Antenna	339A1130AA005
Lightning Arrestor	339A1130AA006
Cable	339A1130AA007
UHF/TNC Adapter	339A1130AA008
Serial Keyboard	339A1130AA009
Electric Line Cord	339A1130AA010
Keyboard Extension Cable	339A1130AA011
Cable	36A357086NAG02
ECMS OIT	36A362595AAG01
Assembly Diagram	36B605697AB
Operator Desk Outline Drwg.	942B508XE

"RMAT" Crane End Replacement Parts

Description	Part Number
Elementary Diagram	255B2612AA001
Elementary Diagram	255B2612AA002
Wireless Modem	339A1130AA003
Power Supply	339A1130AA004
Antenna	339A1130AA005
Lightning Arrestor	339A1130AA006
Cable	339A1130AA007
UHF/TNC Adapter	339A1130AA008
Antenna	339A1131AA001
Cable	36A357086PAG02
Assembly Diagram	36A359027AB



GE Drive Systems

General Electric Company
GE Drive Systems
1100 Lawrence Parkway
Erie, PA 16531

GEK-45153 (1/93)